

Rouaa Guesmi

Data Scientist | Machine Learning, Data Analytics & AI

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SUMMARY

Data Scientist / AI Engineer with an Engineering Degree in Software Engineering, specialized in Data Science, and a 4.0 GPA. Hands-on experience building data pipelines, machine learning and deep learning solutions, predictive/analytical systems, and production APIs across retail, fintech, document intelligence, and computer vision. Strong in Python, SQL/NoSQL, statistical data analysis, ETL, anomaly detection, model evaluation, TensorFlow/PyTorch, Docker, Airflow, MLflow, and cloud environments. Brings an additional edge in Generative AI, RAG, and LLM fine-tuning.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, R, TypeScript, Java, C++, PostgreSQL, MongoDB, MySQL, ETL, Data Cleaning, Data Validation, Feature Engineering
- **Machine Learning & Statistics:** Scikit-learn, TensorFlow, PyTorch, JAX, Statistical Analysis, Predictive Modeling, Classification, Ensemble Modeling, Anomaly Detection, Model Evaluation, Reinforcement Learning
- **Deep Learning & Computer Vision:** 3D CNNs, U-Net, CBAM Attention, ResNet3D, DenseNet3D, OCR, Image Classification, Image Segmentation, NLP
- **Data Engineering & Analytics:** Apache Airflow, Web Scraping, Data Pipelines, PostgreSQL, MongoDB, Tableau, Power BI, Data Quality, Deduplication, GDPR-compliant Anonymization
- **Generative AI:** RAG, Hybrid Retrieval (BM25 + Dense), LoRA/QLoRA, Hugging Face Transformers, LangChain, LLaMA3, Embeddings, LLM Evaluation, AI Agents
- **Cloud, MLOps & Architecture:** Docker, Kubernetes, MLflow, GitHub Actions, AWS, Azure, GCP, FastAPI, Flask, Django, REST APIs, Microservices, Redis, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

- Timsoft Group** | AI Engineer Jan 2026 – Jul 2026
- Built a hybrid retrieval pipeline combining BM25 and dense embeddings with ChromaDB for a Cegid Retail documentation assistant, improving top-5 retrieval accuracy by 12 points over a keyword-only baseline on a 200-query evaluation set.
 - Fine-tuned a 7B-parameter language model with QLoRA on domain Q&A data, improving answer precision on a held-out validation set and reducing unsupported-claim responses versus the base model.
 - Designed multi-step data/AI workflows with tool calling and persistent memory, translating business requirements into automated analytical and support workflows.

- OOLab** | AI Engineering Intern Jul 2025 – Sep 2025
- Contributed to a multilingual, multimodal chatbot processing concurrent text and image inputs across a document-interaction pilot spanning several thousand user sessions.
 - Built an asynchronous document-processing pipeline with Flask and Angular, reducing end-to-end latency versus the previous synchronous flow in internal benchmarks.

- Pixemantic** | FinTech AI Intern Jun 2025 – Aug 2025
- Contributed to PiXCARD, a B2B credit-scoring platform, within Dockerized Django/Flask/FastAPI microservices with RBAC-secured APIs and documented analytical services.
 - Built OCR/NLP models to extract structured fields from financial documents, achieving high extraction accuracy on a labeled internal test set and reducing manual review effort.

- Navinspire IA** | Data Science Intern Jun 2024 – Aug 2024
- Built web-scraping and Apache Airflow ETL pipelines for downstream ML dataset construction, including GDPR-compliant anonymization, deduplication, data-quality checks, and anomaly-detection rules.

- Tunisie Telecom** | Software Engineer Intern Jun 2023 – Aug 2023
- Built a rule-based internship-matching system using .NET Core and C#, automating administrative workflows and matching students with technical supervisors.
 - Designed a microservices architecture and REST APIs, strengthening scalability, maintainability, and integration between application components.

SELECTED DATA SCIENCE & AI PROJECTS

- FinLLM** | Production RAG & MLOps Platform Apr 2025 – Jun 2025
- Built a LLaMA3 + ChromaDB RAG system for financial-document Q&A; created a custom evaluation set and tracked accuracy and latency with MLflow.
 - Optimized analytical-query throughput with FastAPI, Docker, and Redis caching; monitored latency and error rates using Prometheus/Grafana.

- PneumaTect** | 3D Pulmonary Nodule Detection – Capstone Mar 2025 – May 2025
- Co-developed a deep learning pipeline on the public LIDC-IDRI dataset using 3D U-Net with CBAM attention and a ResNet3D/DenseNet3D ensemble for pulmonary nodule detection and classification.
 - Built a Flask inference API and DICOM viewer, connecting model outputs to a usable analytical workflow.

- AutoGPT-Style Planning Agent & MCP Platform** | Autonomous AI / Workflow Automation May 2025 – Nov 2025
- Built a multi-step planning agent with GPT-4 APIs, Redis-backed memory, modular tools, and an MCP integration layer; automated recurring workflows with n8n and Docker.

EDUCATION

- ESPRIT – Private Higher School of Engineering and Technology**, Ariana, Tunisia Sep 2022 – Jun 2026
Engineering Degree in Software Engineering, Specialization in Data Science | GPA: 4.0
- IPEIEM – Preparatory Institute for Engineering Studies El Manar**, Tunis, Tunisia Sep 2020 – Jun 2022
Preparatory Cycle in Mathematics, Physics & Statistics

CERTIFICATIONS

- **NVIDIA Deep Learning Institute:** Building AI Agents with Multimodal Models; Transformer-Based NLP Applications; Fundamentals of Deep Learning; Applications of AI for Anomaly Detection; RAG (Retrieval Augmented Generation)
- **Cloud & Development:** AWS Cloud Foundations (AWS Academy); GitHub Foundations
- **Data & AI:** Data Scientist Professional Track; Python Developer Professional Track; Deep Learning Certification (DataCamp)

LANGUAGES

French (C1) | English (C1) | German (B2)